A sharper polynomial learning bound for NNBayes on robust networks

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Abstract

We study neural-network Bayesian learning (NNBayes) with tanh networks whose clipped predictions are stable under a calibrated ball of simultaneous internal perturbations. Conditioning on positive training margins yields small posterior-averaged test-margin error after polynomially many examples. For a Boolean target represented by a layered circuit of width C, the sample bound is $O(C^6 \log^{13}(2+C))$ when the remaining parameters are fixed. One universal polynomial also controls its dependence on input dimension, depth, robustness calibration, accuracy and confidence; a conservative envelope has degree 46. The result holds at every learner width $N \geq C$ and every Gaussian scale from the finite mean-field prior to stronger priors. It preserves the calibrated finite-perturbation restriction of the polynomial-depth theorem cited below. The improvement comes from importance sampling with a uniform component, exact empirical whitening, and a mean-square comparison of Gaussian residual variances. We include the model, target-mass construction, entropy bookkeeping and statistical proof so that the result can be read independently.

1 Introduction

For Bayesian neural-network learning, a useful statistical guarantee should depend on the complexity of the target rather than on the learner's possibly much larger width. A previous polynomial-depth result obtains such a guarantee after restricting the prior to networks stable under specified internal disturbances [1]. Its direct constructive budget has a very large polynomial degree. This paper improves that budget while leaving the learner and its support restriction unchanged.

The target is a Boolean function computed by a layered bounded-fan-in Boolean circuit. For every learner width above that circuit's width, conditioning the restricted Gaussian prior on exact training margins gives small posterior test error after a number of examples polynomial jointly in circuit width, depth and the other parameters. We establish an explicit sixth-power dependence on circuit width, up to logarithms, including at finite mean-field scale. Quadratic dependence under this same restriction remains unresolved.

The main analytic issue is that a wide mean-field network contains a large Gaussian bulk whose Frobenius energy grows with width. Simply retaining low-energy networks does not yield a width-independent bound. We retain the previous entropy-based separation into informative directions and a Gaussian residual, but reconstruct that residual more accurately. Sampling from a mixture of uniform and leverage probabilities controls both active and residual errors. Exact

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whitening makes the sampled covariance a least-squares residual; comparing square roots of its variances then costs $O(q/m)$ for q active directions and m sampled neurons. This replaces a much less efficient uniform covariance approximation.

Status and scope. The headline learning theorem, the sixth-power width rate and the degree-46 envelope have Lean proofs. The separate numerical allocation in the explicit-budget section is outside that certification. Independent AI mathematical reviews are also available through the formal-coverage record. The formal circuit class reserves a counted constant-$+1$ wire at every stage, including the final stage. In particular, a width-one circuit other than constant $+1$ under the prose convention below needs formal width two, so its learner-width-one case is not certified. The verified width threshold uses the formal circuit width. The robustness assumption is a restriction on supported implementations, not a claim about all gradient-trained networks. Targets are layered Boolean circuits, rather than arbitrary real-weight neural circuits of the same width. The guarantee concerns exact Bayesian inference and does not provide an efficient posterior-sampling algorithm.

Reading guide. Section 2 defines the Gaussian prior, margins, and finite-perturbation support. Section 3 states the learning guarantee. Its proof in Section 4 combines a finite approximation class with the target-neighborhood probability estimate of Section 5. Section 6 gives the numerical allocation and derives the polynomial exponents. Appendix A constructs the finite approximation class, including the whitened sampling step that improves the width dependence.

2 Model, prediction convention, and stability restriction

Let $\mathcal{X} = \{-1, 1\}^d$, let D be any distribution on $\mathcal{X}$, and let $f : \mathcal{X} \to \{-1, 1\}$ be the target. A network has $H = L - 1$ hidden layers, each of width N, tanh activations, and a linear readout:

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}) \ (2 \le s \le H), \qquad z_w(x) = v^\top h^H, \qquad \bar{x} = (x, 1).$$

The first layer includes a bias coordinate; internal constants are carried by ordinary hidden wires. Let $\mu_{N,t}$ be the product Gaussian with entry variances t^2/d in W^1, t^2/N in the internal matrices, and t^2/N^2 in v, where $0 < t \le 1$. The exponent convention $t = N^{(1-\gamma)/2}$ therefore gives every $\gamma \ge 1$. Define

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N\|v\|^2, \qquad E_v(w) = N\|v\|^2.$$

Here $\|\cdot\|_F$ is the Frobenius norm, the square root of the sum of squared matrix entries. Vector norms are Euclidean, and $\|\cdot\|_{\mathrm{op}}$ below is the corresponding operator norm. The scale $t = 1$ is the finite mean-field prior; smaller positive t concentrates its weights more strongly near zero. In these energy coordinates the Gaussian density is proportional to $\exp[-N\kappa_N/(2t^2)]$.

A disturbance is added after every hidden activation and at the readout:

$$h^1_\eta = \tanh(W^1 \bar{x}) + \eta^1, \qquad h^s_\eta = \tanh(W^s h^{s-1}_\eta) + \eta^s, \qquad z_w(x; \eta) = v^\top h^H_\eta + \eta^{\mathrm{out}}.$$

The simultaneous disturbance budget is

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \frac{\|\eta^s\|_2^2}{N} + |\eta^{\mathrm{out}}|^2. \tag{1}$$

This is an adversarial finite perturbation, not a Gaussian-noise average or a derivative evaluated only at the clean computation. Noisy hidden states need not stay in $[-1, 1]^N$.

Draw the training inputs independently from D, so the sample is independent and identically distributed (iid). Our training likelihood is

$$\mathcal{L}_S(w) = \prod_{i=1}^{n} \mathbf{1}\{f(X_i)z_w(X_i) > 1/2\}. \tag{2}$$

At test time we count $f(x)z_w(x) \le 1/4$ as an error. This stronger loss also bounds sign error. There is no requirement that unobserved inputs have any clean output margin.

Clip the output at the training margin:

$$T(u) = \max\{-1/2, \min\{1/2, u\}\}.$$

For a calibration parameter $c > 0$, set

$$r_c(w) = \frac{c}{\sqrt{1 + HE_v(w)}}. \tag{3}$$

Let $\mathcal{R}_{N,c}$ be the set of networks satisfying

$$|T(z_w(x;\eta)) - T(z_w(x))| \le 1/16 \quad \text{for every } x \in \mathcal{X} \text{ and every } \|\eta\|_* \le r_c(w). \tag{4}$$

The prior is exactly $\mu_{N,t}$ conditioned on $\mathcal{R}_{N,c}$. The posterior $Q_{S,N,t,c}$ additionally conditions on (2). Both conditioning events have positive probability by the target construction below. No tractable sampling algorithm is assumed or supplied by the theorem.

The clipping scale is set by the training margin in (2). For example, changing a raw output from 10 to 1 causes no score change. Clipping at $\pm 1/2$ gives a weaker condition than clipping at ± 1 with the same tolerance and noise radius, because $T = T \circ T_1$, where T_1 clips at ± 1, and T is 1-Lipschitz.

3 The learning theorem

A width-C, depth-H layered Boolean circuit has at most C Boolean wires at each of its H stages. We use NAND gates, copies, and negations; constant wires required by internal gate implementations are carried through the circuit and count toward C. This is a universal Boolean basis. Compiling another fixed basis into this one may change the width and depth, and those resources must be counted. Its output is one final wire. The circuit's logical depth must fit the H hidden layers; idle stages can be filled with copies.

Theorem 3.1 (Polynomial depth at every width and every $\gamma \ge 1$)**.** *There is a universal polynomial P with the following property. The explicit sufficient sample budget is given in Section 6. At fixed $d, L, c, \varepsilon, \delta$, it is $O(C^6 \log^{13}(2+C))$. Let f be computed by such a width-C, depth-H Boolean circuit, and let D be any input distribution. For every $c > 0$ and every $0 < \varepsilon, \delta < 1$, if*

$$n \ge P\big(d, L, C, c + c^{-1}, \varepsilon^{-1}, \log(1/\delta)\big),$$

then with probability at least $1 - \delta$ over the iid labeled sample,

$$\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X)z_w(X) \le 1/4\} \le \varepsilon \tag{5}$$

simultaneously for every $N \ge C$ and $0 < t \le 1$. The polynomial degree is independent of width, depth, and Gaussian scale. In particular, the result includes the actual finite-variance mean-field scale $\gamma = 1$ and every stronger-prior scale $\gamma > 1$.

The same theorem holds if the prior is additionally conditioned on $|z_w(x)| > 1/2$ at every input. This is the globally margin-conditioned variant discussed in [1]. The proof below applies unchanged to this variant. The target cylinder constructed below already satisfies this extra condition. Its probability bound and every subsequent statistical step are therefore unchanged. Under that additional condition, test-margin error coincides with sign error. Dropping the extra condition is permitted, but is not required for the result.

The width threshold concerns a layered Boolean circuit, not an arbitrary real-valued neural-network representation of the same Boolean function. The theorem does not assert that every function with a small real neural circuit has a comparably small Boolean circuit at the same depth, or that all its low-energy neural representations satisfy the stability condition. The proof instead supplies one robust implementation of each Boolean circuit, together with enough Gaussian probability around that implementation.

Define the posterior correct-margin probability $p_S(x) = Q_{S,N,t,c}\{f(x)z_w(x) > 1/4\}$. Equation (5) immediately implies

$$D\{x : p_S(x) \le 2/3\} \le 3\varepsilon, \qquad D\{x : p_S(x) \le 1/2\} \le 2\varepsilon.$$

The same bounds hold for posterior sign correctness, since the correct-sign event contains the correct-margin event. These conclusions allow arbitrary posterior mixtures; they do not require concentration on one predictor.

4 Compression and the statistical proof

The constructive compression statement needed here concerns numerical outputs before any robustness condition is used. We write $D_{\mathrm{KL}}(Q\|\mu)$ for the Kullback–Leibler (KL) divergence, or relative entropy, of Q from μ. The common law Γ_Q in the next lemma lets the statistical proof average over one finite class even though the comparison coupling can vary with the input. The notation $Q \ll \mu$ means that Q is absolutely continuous with respect to μ.

Lemma 4.1 (Entropy compression before the prediction comparison)**.** *For $d \ge 1$, $L \ge 2$, $K \ge 1$, $0 < r \le 1$, and $0 < \alpha \le 1/8$, there is a finite class $\mathcal{H}$ of real-valued bounded-weight depth-L tanh networks, independent of N, t, such that*

$$\log|\mathcal{H}| \le \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

If $Q \ll \mu_{N,t}$ satisfies $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$, there is one distribution Γ_Q on $\mathcal{H}$ with the following property. For every fixed x, one can couple $w \sim Q$ and $h \sim \Gamma_Q$ so that, except with probability at most α, there is a disturbance η with

$$\|\eta\|_* < r/2, \qquad |h(x) - z_w(x;\eta)| \le r/8.$$

The couplings and disturbances may depend on x, but their two marginals are the same at every input. No robustness or clean-margin assumption is made on Q.

The proof is constructive: it retains a bounded number of input directions carrying conditional Gaussian relative entropy in each layer, replaces the remaining directions by Gaussian randomness, samples a bounded number of source neurons, and rounds a bounded-size network to a fixed grid. The resulting virtual computation is the original network under small postactivation disturbances. The appendix gives the complete construction; the prediction comparison below is the first step that uses stability.

Lemma 4.2 (Readout truncation)**.** *Suppose a posterior is the raw Gaussian conditioned on an event of mass at least* $\exp(-ND_*/t^2)$*, where* $D_* \geq 1$*. For* $0 < \beta \leq 1/2$*, put*

$$B = \max\{8, 8[D_* + \log(1/\beta)]\}.$$

It assigns mass at most β *to* $E_v > B$*. Its conditional law* Q' *on* $E_v \leq B$ *satisfies* $D_{\mathrm{KL}}(Q'\|\mu_{N,t}) \leq N(D_* + 1)/t^2$*.*

Proof. Under the Gaussian, NE_v/t^2 has the chi-square distribution with N degrees of freedom. Its exponential moment gives, for $B \geq 8$,

$$\mu_{N,t}\{E_v > B\} \leq e^{-NB/(4t^2)}2^{N/2} \leq e^{-NB/(8t^2)}.$$

Divide by the conditioning-event probability and use $N/t^2 \geq 1$. The further conditioning event has raw probability at least $(1-\beta)e^{-ND_*/t^2}$, so its negative logarithm is at most $ND_*/t^2 + \log[1/(1-\beta)] \leq N(D_* + 1)/t^2$. □

Statistical proof of Theorem 3.1. The next section constructs a subset of the support with correct noisy margin on every input and raw mass at least e^{-ND_*/t^2}, where $D_* = \mathrm{poly}(d, L, C, c)$ is independent of N, t. It is contained in the posterior conditioning event for every correctly labeled sample. Apply Lemma 4.2 with $\beta = \varepsilon/4$, and write Q' for the retained law. Set

$$K = D_* + 1, \qquad r = \min\{1/6, c/\sqrt{1 + HB}\}, \qquad \alpha = \varepsilon/12.$$

Every retained network satisfies the score-stability condition at the common radius r. Lemma 4.1 and the 1-Lipschitz property of T therefore give pointwise couplings with

$$\Pr\{|T(h(x)) - T(z_w(x))| > 1/12\} \leq \alpha,$$

because $1/16 + r/8 \leq 1/12$.

Use the bounded ramp loss

$$\psi(u) = \begin{cases} 1, & u \leq 1/3, \\ 5 - 12u, & 1/3 < u < 5/12, \\ 0, & u \geq 5/12. \end{cases}$$

At a training input, $f(x)z_w(x) > 1/2$ implies $f(x)T(z_w(x)) = 1/2$. A successful coupling therefore gives $f(x)T(h(x)) \geq 5/12$, so its ramp loss is zero. The candidate's averaged empirical ramp risk is at most α. At a test input with $f(x)z_w(x) \leq 1/4$, a successful coupling gives $f(x)T(h(x)) \leq 1/3$, so its ramp loss is one.

For any finite class of $[0,1]$-valued losses, a relative Chernoff bound and a union bound give, with probability at least $1 - \delta$,

$$\mathbb{E}_D\psi(f(X)T(h(X))) \leq \frac{2}{n}\sum_{i=1}^{n}\psi(f(X_i)T(h(X_i))) + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n} \quad (h \in \mathcal{H}).$$

For completeness, $e^{-u} \leq 1 - (1 - e^{-1})u$ on $[0,1]$ gives the usual lower-tail exponent at least the mean divided by eight. This proves the displayed bound for one loss, and a union bound proves simultaneity. Average under the sample-dependent law $\Gamma_{Q'}$, apply the population comparison, and restore the discarded posterior mass. The result is

$$\Pr_{Q_{S,N,t,c},D}\{f(X)z_w(X) \leq 1/4\} \leq \beta + 3\alpha + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n}.$$

Choose n so the last term is at most $\varepsilon/2$. Every quantity entering $\log|\mathcal{H}|$ is bounded by a universal polynomial in the theorem's parameters. Its class and the displayed sample event are independent of N, t, establishing the simultaneous statement. □

5 A robust target cylinder at every width

We give the probability bound used in the statistical proof. It uses a robust digital implementation, not a claim that every implementation is stable. Let $N \geq C$. Partition the neurons into C balanced clone groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$, so $C/2 \leq N/k_a \leq 2C$. Let P project onto vectors constant on each clone group. Fill unused logical slots with available constants or copies of an existing Boolean wire. The ideal wire vector $b^s(x)$ then has entries in $\{-1, 1\}$, belongs to the range of P, and has Euclidean norm $\sqrt{N}$.

Use gain A in the tanh simulation of every Boolean gate. For example, NAND on $u, v \in \{-1, 1\}$ is the sign of $1 - u - v$; copies and negations use u and $-u$. Bias constants in internal layers are supplied by counted constant wires. The ideal signed preactivation of every gate is at least A. In the clone implementation, each coefficient from a logical wire is divided equally among that wire's clones. Set

$$e_* = \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = \frac{e_*}{4}, \qquad b_L = \sqrt{16 \log 9 + 8 \log(4L)},$$
$$A = \max\{32 b_L, 2 \log(8/e_*)\}, \quad a = \max\{2, 256c\}, \quad \rho = \min\{1/4, A/[4(d+1)], A e_*/(64\sqrt{C})\}.$$

The teacher readout averages the clones of the output wire with total coefficient a.

In Gaussian coordinates $\sqrt{d}\, W^1$, $\sqrt{N}\, W^s$, and Nv, all entries have variance t^2. Restrict each first-layer coordinate, each internal row's projection onto every normalized clone-group direction, and each readout coordinate to an interval of radius ρt about its teacher value. The internal matrices then decompose as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \leq \rho t \sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk E^s is independent of the constrained coordinates. A $1/4$-net of the unit sphere has at most 9^N points, and the usual two-net Gaussian bound gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > b_L t\} \leq 2 \exp(2N \log 9 - N b_L^2/8) \leq \frac{1}{2L}.$$

Thus all bulk norms are at most $b_L t$ with conditional probability at least $1/2$.

Finite disturbance robustness. Allow a separate root-mean-square (RMS) disturbance of size τ_0 at each hidden layer and scalar output disturbance at most τ_0. This includes the entire total-budget ball $\|\eta\|_* \leq \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \leq e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_a} \leq 1/16$. At most three teacher coefficients, including a constant coefficient, enter a gate, so its intended signed preactivation remains at least $A/2$. For $\epsilon \in \{-1, 1\}$ and real u, the elementary global secant bound

$$\epsilon q \geq A/2 \quad \Longrightarrow \quad |\tanh(q + u) - \epsilon| \leq 2e^{-A/2} + 8|u|/A$$

holds: for $|u| < A/4$ use saturation, and otherwise the linear term is at least two. Since $E^s b^{s-1} = 0$ and $\|h^{s-1}_\eta\|_2/\sqrt{N} \leq 1 + \tau_0$, it follows that

$$e_s \leq 2e^{-A/2} + (8/A)\rho\sqrt{C}(1 + \tau_0) + (8b_L/A)e_{s-1} + \tau_0 \leq e_*.$$

Each of the four terms is at most $e_*/4$. The first layer initializes the same invariant: its coordinatewise preactivation perturbation is at most $\rho(d+1) \leq A/4$, and its postactivation disturbance contributes at most τ_0 in RMS norm. Consequently the noisy signed readout is at least

$$f(x) z_w(x; \eta) \geq \frac{15a}{16} - \rho(1 + \tau_0) - \tau_0 > 1 \qquad (x \in \mathcal{X},\ \|\eta\|_* \leq \tau_0).$$

The readout center satisfies $\sqrt{N}\|v_*\| \geq a\sqrt{C/2}$, and its permitted perturbation has $\sqrt{N}\|v-v_*\| \leq \rho$. Therefore $E_v \geq a^2C/8$, whence

$$r_c(w) \leq \frac{2\sqrt{2}c}{a\sqrt{C}} \leq \frac{1}{64\sqrt{2C}} = \tau_0.$$

All clean and noisy clipped scores on the calibrated ball equal $f(x)/2$. The entire cylinder, with the bulk-norm event imposed, therefore belongs to $\mathcal{R}_{N,c}$ and satisfies the training likelihood for every sample. This also proves why every fixed $c > 0$ is allowed: a larger calibration requires only a polynomially larger readout coefficient.

Gaussian mass. There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. The sum of their squared centers is at most NB_0, with

$$B_0 = 9dA^2 + 18(L-2)CA^2 + 2a^2C.$$

These loose constants cover the fan-in-two gate implementation above. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \leq 1/4$, and any center m,

$$\Pr\{|Z - m| \leq \rho t\} \geq \frac{2\rho}{e\sqrt{2\pi}} \exp(-m^2/t^2).$$

Multiplication over the coordinates and the independent bulk event gives

$$\mu_{N,t}(\text{target cylinder}) \geq \exp(-ND_*/t^2), \qquad D_* = B_0 + q \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2. \tag{6}$$

Here $D_* \geq 1$ and is bounded by a universal polynomial in d, L, C, c. The construction and estimate hold at every integer width $N \geq C$.

6 Explicit sample budget and dependence on circuit width

The following formulas are a sufficient bound, with no implicit dependence on learner width or Gaussian scale. They are the manuscript's numerical allocation; Lean certifies the headline polynomial bounds and exponents, but not these exact constants. Given the target-cylinder cost D_* from (6), put

$$\beta = \varepsilon/4, \qquad B_v = \max\{8, 8[D_* + \log(1/\beta)]\}, \qquad K = D_* + 1,$$
$$r = \min\{1/6, c/\sqrt{1 + (L-1)B_v}\}, \qquad \alpha = \varepsilon/12.$$

Use these K, r, α in the parameter choices of Appendix A, which defines N_0, B_{small}, and B_{large}. Set $M = N_0$, $B = \max(B_{\text{small}}, B_{\text{large}}, 1)$,

$$P = M(d+1) + (L-2)M^2 + M, \qquad J = P \log\left[3 + \frac{32B(d+L+1)(2M(B+1))^L}{r}\right].$$

Then the explicit sufficient sample size is

$$n \geq \frac{16}{\varepsilon}\left[J + \log\frac{1}{\delta}\right]. \tag{7}$$

The statistical proof above is unchanged; only the value of J improves.

Circuit width. For fixed $d, L, c, \varepsilon, \delta$, the target construction gives $D_* = O(C \log^2(2 + C))$. Write $F = C \log^2(2 + C)$. The budgets satisfy

$$K, r^{-2} = O(F), \quad q, V = O(F^2), \quad m, M = O(F^3), \quad B = O(F^3), \quad P = O(F^6).$$

The logarithmic grid cost is $O(\log(2 + C))$ per parameter. Consequently

$$n = O(C^6 \log^{13}(2 + C)).$$

This holds for the actual finite mean-field prior as well as every stronger scale, uniformly over all learner widths $N \geq C$. It is an upper bound; the argument does not show that the sixth power is necessary.

A conservative universal polynomial. Set

$$S = 2 + d + L + C + c + c^{-1} + \varepsilon^{-1} + \log(1/\delta).$$

For universal constants, each quantity below is $O(S^p)$ with the displayed exponent. Logarithms are absorbed only where indicated.

Quantity	p
D_*, K, B_v	3
r^{-2}	6
e^{-1}, η^{-1}	8, 2
q, V	13
D_0	14
m, M	21
P	43
B^2_{small}, B	44, 22

For example the reconstruction term $(V + q + 1)/e$ is $O(S^{21})$, the Gram term is $O(S^{14})$, and the readout term is $O(S^{12})$. The grid uses $O(S \log S)$ bits per coordinate. Thus $J = O(S^{44} \log S)$ and (7) is $O(S^{45} \log S)$, bounded by $O(S^{46})$. This degree is a conservative multivariate envelope, not a claim of a practical numerical guarantee. The headline theorem and degree-46 envelope now have Lean proofs, as does the improved compressor in the form the main theorem uses (for tanh, and without the robustness assumption only at large width). The particular numerical allocation in this section, including its leading factor $16/\varepsilon$, is outside that certification.

Remaining quadratic question. The calibrated finite-perturbation support (4) is preserved throughout this paper. It remains open here whether its sample bound can be quadratic in target circuit width. The current method retains $q = O(F^2)$ active directions and samples $m = O(F^3)$ source neurons; coding the dense compressed network then costs $O(F^6)$ parameters. A quadratic conclusion would require a further improvement, rather than a reinterpretation of this bound.

A Finite posterior compression by internal-noise coupling

This appendix gives the improved compression construction used in the main theorem. It tracks relative entropy only in the original weights that have not yet been processed. The compressed

prefix need not have a Gaussian density. No stability assumption is needed until the coupling is used in the main statistical argument. The theorem below states the manuscript's general activation version. The Lean coverage is narrower: it certifies the tanh compressor, and the version without a robustness assumption only at large width. Those are the cases used in the formal proof of the main learning theorem.

Theorem A.1 (Finite compression before invoking stability)**.** *Fix a scalar* 1*-Lipschitz activation* $|\phi| \le 1$*, and parameters* $d \ge 1$*,* $L \ge 2$*,* $K \ge 1$*,* $0 < r \le 1$*, and* $0 < \alpha \le 1/8$*. There is a finite class* $\mathcal{H} = \mathcal{H}(d, L, K, r, \alpha)$ *of real-valued bounded-weight depth-*L *network outputs with*

$$\log |\mathcal{H}| \le \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

The polynomial and class are independent of width N *and scale* $0 < t \le 1$*. For every* $Q \ll \mu_{N,t}$ *with* $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$*, there is a law* Γ_Q *on* $\mathcal{H}$ *such that, at every Boolean input* x*, one can couple* $w \sim Q$*,* $h \sim \Gamma_Q$*, and a disturbance* η *so that*

$$\mathbb{P}\left\{\|\eta\|_* < r/2, \quad |h(x) - z_w(x;\eta)| \le r/8\right\} \ge 1 - \alpha.$$

The coupling may depend on x*; its two specified marginals do not.*

We give the proof with explicit budgets. The entropy bookkeeping and active-direction lemma below are the ingredients from [1]; their statements and proofs are repeated here. The subsequent sampling lemma sharpens the reconstruction step. Two elementary entropy facts organize the construction. First, for a standard D-dimensional Gaussian G, the entropy variational inequality gives

$$\mathbb{E}_P\|Z\|^2 \le 4D_{\mathrm{KL}}(P\|G) + 2D\log 2. \tag{8}$$

Second, Gaussian Talagrand [2] gives $W_2^2(P, G) \le 2D_{\mathrm{KL}}(P\|G)$, where W_2 is the quadratic Wasserstein distance, obtained by minimizing expected squared Euclidean distance over couplings and taking a square root. In the next lemma, $H(\cdot \mid \cdot)$ denotes conditional Shannon entropy and I denotes mutual information. All logarithms and entropies use natural units.

Lemma A.2 (Entropy of the unprocessed weights)**.** *Let* A, X, Y *be random variables, with fixed reference laws* P_X, P_Y*, and put*

$$B = \mathbb{E}_A D_{\mathrm{KL}}(Q_{X,Y|A}\|P_X \otimes P_Y).$$

Let J *be a finite message, possibly depending on* (A, X, Y)*, and let* $H = f(A, J, X)$*. If* A' *is a randomized function of* (A, J, H) *whose fresh randomness is conditionally independent of* Y*, then*

$$\mathbb{E}_{A'} D_{\mathrm{KL}}(Q_{Y|A'}\|P_Y) \le B + H(J \mid A).$$

Proof. The left side equals $D_{\mathrm{KL}}(Q_Y\|P_Y) + I(Y; A')$. Data processing and

$$I(Y; J, H \mid A) \le I(Y; X, J \mid A) \le I(Y; X \mid A) + H(J \mid A)$$

bound it by $\mathbb{E}_{A,X} D_{\mathrm{KL}}(Q_{Y|A,X}\|P_Y) + H(J \mid A)$, which is at most the claimed bound by the KL chain rule. These expressions can equivalently be justified by finite partitions and lower semicontinuity. □

Here A will be a constructed prefix, X the current original weight block, and Y all later original blocks. The lemma imposes no reference law on A.

Lemma A.3 (Whole-layer active directions)**.** *Condition on (A, Y) and let X be a standardized Gaussian-reference matrix, with conditional relative entropy $R(A, Y)$. For vectors $b_x(A)$ of norm at most one, one can select a list J of inputs and its span $U = \operatorname{span}\{b_x : x \in J\}$ with*

$$\dim U \le \frac{t^2 R(A, Y)}{N\delta}$$

such that preserving $H = tXU$ and replacing the orthogonal coordinates by fresh Gaussian coordinates changes the physical evaluation vectors tXb_x, under a suitable coupling, by expected squared RMS at most 2δ at each x. The candidate matrix law preserves (A, Y, H) and does not depend on x.

Proof. Use a fixed ordering of the finite input set. Whenever the averaged conditional KL of the entire residual evaluation vector Xv exceeds $N\delta/t^2$, add the unit residual direction v of that input. Each query is selected from the conditional law given (A, Y), averaging over active-coordinate values; thus J is a function of (A, Y). Orthogonal Gaussian coordinates are independent, so the active-coordinate entropy increases by precisely this conditional KL. The displayed rank bound follows. At termination, Gaussian Talagrand on the entire evaluation vector, multiplication by the squared residual-vector length, and division by N/t^2 give the error bound. Scalar-vector couplings extend to full matrices by disintegration. The basis of U is a measurable function of A, J under fixed Gram–Schmidt; it has no further dependence on Y once those variables are supplied. □

Lemma A.4 (Whitened sampling with a uniform component)**.** *Suppose $|h_j(x)| \le 1$ for $1 \le j \le N$, and U has $\ell \le q$ orthonormal columns. Put $u_j = \sqrt{N}\, U_{j,:}$,*

$$\psi(x) = N^{-1} \sum_j u_j h_j(x), \quad K(x, y) = N^{-1} \sum_j h_j(x) h_j(y), \quad C = K - \psi^\top \psi.$$

For $\ell > 0$ sample independently with

$$p_j = \frac{1 + \|u_j\|^2/\ell}{2N}, \qquad a_j = \frac{1}{N p_j}.$$

For $\ell = 0$ use uniform sampling, $a_j = 1$, and empty active matrices. For m sampled indices J_b define

$$\widehat{G} = \frac{1}{m} \sum_b a_{J_b} u_{J_b} u_{J_b}^\top, \quad \widehat{v}(x) = \frac{1}{m} \sum_b a_{J_b} u_{J_b} h_{J_b}(x), \quad \widehat{K}(x, y) = \frac{1}{m} \sum_b a_{J_b} h_{J_b}(x) h_{J_b}(y).$$

Except on an event of probability at most $\ell e^{-m/[16(\ell+1)]}$, $\widehat{G} \succeq I/2$. On this event set

$$D_{b,:} = \sqrt{a_{J_b}/m}\, u_{J_b}^\top \widehat{G}^{-1/2}, \qquad \xi_i \sim \mathcal{N}(0, I - DD^\top),$$

independently over target rows i. For any active matrix H, the rows

$$\widehat{w}_{ib} = \frac{a_{J_b}}{m} H_i \cdot u_{J_b} + t\sqrt{\frac{a_{J_b}}{m}}\, \xi_{ib} \tag{9}$$

have one input-independent law. At each fixed input their evaluations can be coupled to the ideal field with mean $H\psi(x)$ and independent row Gaussian variance $t^2 C(x, x)$ so that the expected squared RMS error, with the good-event indicator, is at most

$$\frac{2\|H\|_F^2/N + 16(q+1)}{m}. \tag{W}$$

Proof. The source law is a half-uniform, half-leverage mixture; hence $a_j \le 2$ and $a_j\|u_j\|^2 \le 2\ell$. The summands in $\widehat{G}$ are positive semidefinite, bounded by $2\ell I$, and have expectation I. The matrix Chernoff lower-tail inequality [3] gives the stated conservative failure bound. All empty-matrix claims are immediate for $\ell = 0$.

On success $D^\top D = I$, so the Gaussian construction is valid. Its mean is $H\widehat{v}(x)$ and covariance is

$$t^2\widehat{C}(x,y), \qquad \widehat{C} = \widehat{K} - \widehat{v}^\top \widehat{G}^{-1}\widehat{v}.$$

Equivalently its active coefficient is $H\widehat{G}^{1/2}$ against the whitened features; omitting this square-root compensation would give a different estimator. The active estimator is unbiased and

$$\mathbb{E}\frac{\|H(\widehat{v}(x) - \psi(x))\|^2}{N} \le \frac{2\|H\|_F^2}{Nm},$$

because its covariance is bounded by $m^{-1}\mathbb{E}_{j\,\mathrm{uniform}} a_j u_j u_j^\top h_j(x)^2 \preceq 2I/m$.

For the Gaussian part fix x, abbreviate $C = C(x,x)$, and set

$$R_j = h_j(x) - u_j^\top \psi(x), \quad \widehat{c} = \frac{1}{m}\sum_b a_{J_b} R_{J_b}^2, \quad b = \frac{1}{m}\sum_b a_{J_b} u_{J_b} R_{J_b}.$$

Then $\mathbb{E}_{\mathrm{uniform}} u_j R_j = 0$, $\mathbb{E}_{\mathrm{uniform}} R_j^2 = C \le 1$, and the exact empirical least-squares identity is

$$\widehat{C}(x,x) = \widehat{c} - b^\top \widehat{G}^{-1} b.$$

Since $\|\psi(x)\| \le 1$, $a_j R_j^2 \le 4(\ell+1)$. Thus

$$\mathrm{Var}(\widehat{c}) \le \frac{4(\ell+1)C}{m}, \qquad \mathbb{E}\|b\|^2 \le \frac{2\ell C}{m}.$$

If $C > 0$, rationalizing the square root gives

$$\mathbb{E}(\sqrt{\widehat{c}} - \sqrt{C})^2 \le \frac{\mathbb{E}(\widehat{c} - C)^2}{C} \le \frac{4(\ell+1)}{m}.$$

If $C = 0$ every residual vanishes. On the good Gram event,

$$(\sqrt{\widehat{c}} - \sqrt{\widehat{C}(x,x)})^2 \le b^\top \widehat{G}^{-1} b \le 2\|b\|^2.$$

The squared triangle inequality consequently gives

$$\mathbb{E}[\mathbf{1}_{\mathrm{good}}(\sqrt{\widehat{C}(x,x)} - \sqrt{C})^2] \le \frac{16(\ell+1)}{m}.$$

Use a common standard normal to couple the two scalar Gaussian evaluations. The mean and variance contributions have zero cross term. Adding the active estimate proves (W), since $t \le 1$. The marginal laws are defined for the whole field and do not depend on the input; scalar couplings extend to coefficient fields by disintegration. No simultaneous field coupling is claimed. □

Parameter choices. Put

$$\begin{aligned} e &= \alpha r^2/(1000L), & \eta &= \alpha/(100L), & q &= \lceil 2K/(e\eta)\rceil, \\ D_0 &= [(d+1)q+2]\log 2, & V &= 8K+2q, & B_H^2 &= 4V/e, \\ K_0 &= (4K+2\log 2)/\eta, & B_1^2 &= 4[4K+2(d+1)\log 2]/e. \end{aligned}$$

Let m be the least integer no smaller than all of

$$\begin{aligned} &10000(V+q+1)/e, \\ &64(q+1)\log(4(q+1)/\eta), \\ &512K_0 r^{-2}[d\log 2+\log(2/\eta)+1]. \end{aligned} \tag{10}$$

Finally put

$$N_0 = \max\{m, \lceil LD_0/K\rceil, 1\}, \qquad B_G = \max\{2, 2\sqrt{\log(6m/e)}\}.$$

All quantities have a fixed polynomial bound in the theorem parameters.

The small-width case. If $N \le N_0$, let $P_0 = N_0(d+1)+(L-2)N_0^2+N_0$. Applying (8) to all standardized weights, and then multiplying by t^2, gives

$$\mathbb{E}_Q\|tZ_{\text{all}}\|^2 \le 4N_0K + 2P_0\log 2.$$

With probability at least $1-\alpha/4$, all actual weights are bounded by

$$B_{\text{small}} = [4(4N_0K+2P_0\log 2)/\alpha]^{1/2}.$$

Keep the original network on this event and use a default zero-readout network otherwise. Padding with nodes having zero outgoing weights preserves the original output. It remains to treat $N > N_0$.

The virtual prefix and its entropy budget. The constructed prefix A has N virtual nodes in its current layer. Each virtual row acts on m stored nodes of the preceding layer. The auxiliary description may keep all N rows; the final network will keep just the rows selected at the following stage. Track the conditional KL of the unprocessed *original* weights. By Lemma A.2, it is at most

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2. \tag{*}$$

This is not a KL bound for the singular compressed prefix itself.

For the first layer, zero every physical standardized row $tZ_{1,i}$ of norm greater than B_1, and retain all other rows. Bounded activations and (8) bound the expected squared RMS activation change at each input by e. This deterministic prefix channel does not increase the future entropy budget.

An internal layer: retain active data. Condition on the current prefix A and the future original weights Y. Apply Lemma A.3 to $b_x = h_A(x)/\sqrt{N}$ and $\delta = e$. Abort if more than q inputs are needed. By (*) and Markov, this failure has probability at most η. The successful query list, together with an abort flag, has a code of at most $[(d+1)q+2]$ bits. Its entropy is at most D_0.

Let $H = tXU$ be the physical active matrix. Averaging the Gaussian entropy moment bound, with the success indicator rather than conditioning on success, gives

$$\mathbb{E}\left[\frac{\|H\|_F^2}{N};\ \text{success}\right] \le 4t^2B_s/N + 2t^2q\log 2 \le V.$$

Zero every row H_i whose norm exceeds B_H. The expected squared RMS activation change from this operation is at most $4V/B_H^2 = e$. All rows are clipped before any later selection, including selections that may depend on future weights.

An internal layer: whitened source reconstruction. Apply Lemma A.4 with the selected active basis and the clipped active matrix. Its probabilities depend only on A, J. Abort on the bad Gram event, of probability at most η by (10). Integrating (W) with the success indicator and using the active-energy bound V gives expected squared RMS reconstruction error at most $e/8$.

Clip every coordinate ξ_{ib} to $[-B_G, B_G]$ before any later row selection. Each has variance at most one, and $a_j \le 2$, so

$$\mathbb{E}\left|\frac{t}{\sqrt{m}}\sum_b \sqrt{a_{J_b}}(\xi_{ib} - \mathrm{clip}_{B_G}\xi_{ib})h_{A,J_b}(x)\right|^2 \le 6me^{-B_G^2/4} \le e.$$

Every coefficient is bounded by

$$2B_H\sqrt{q+1}/m + \sqrt{2}B_G/\sqrt{m},$$

using $a_j\|u_j\| \le 2\sqrt{q+1}$. These bounds hold before selection depending on future original weights. The new prefix is a randomized function of A, J, H, with fresh randomness conditionally independent of the future weights. Gram failure depends only on A, J and the source samples. Lemma A.2 therefore preserves the entropy budget (*), without a new message cost. Entropy replacement, active clipping, reconstruction and Gaussian clipping contribute less than $20e$ expected squared RMS activation injection, by the squared triangle inequality.

The readout and final network. The original readout marginal satisfies $\mathbb{E}_Q[N\|v\|^2] \le 4K + 2\log 2$. Abort if $N\|v\|^2 > K_0$, with probability at most η. Put $A_1 = \sum_i |v_i| \le \sqrt{K_0}$. When $A_1 > 0$, sample final source indices with probabilities $|v_i|/A_1$ and use readout coefficients $A_1\,\mathrm{sign}(v_i)/m$. The zero-readout case is immediate. Conditional on the prefix and readout, these are independent bounded estimates of $v^\top h_A(x)$. Hoeffding and a union bound over the 2^d inputs give error at most $r/16$ everywhere, except with probability η, by the last line of (10). No future block remains, so readout-dependent source probabilities require no entropy charge.

Keep just the m source rows chosen from each virtual layer. Each already acts on the preceding m stored nodes; repeated indices are duplicate neurons. The result is an ordinary width-m network with weight bound

$$B_{\mathrm{large}} = \max\{B_1, 2B_H\sqrt{q+1}/m + \sqrt{2}B_G/\sqrt{m}, \sqrt{K_0}/m, 1\}.$$

The original network receives all perturbations. Fix an input x and glue the conditional couplings, retaining the original weight history. The current and future original blocks are untouched until their own stages. Define

$$\eta^s = h^s_{\mathrm{virtual}}(x) - \phi(W^s_{\mathrm{original}}h^{s-1}_{\mathrm{virtual}}(x)).$$

The analogous readout difference is η^L. Later stages leave earlier computational prefixes unchanged, so the final candidate output is exactly the *original* network's output under these postactivation injections. No intermediate robustness assumption is made.

For precision, the canonical construction draws its new Gaussian coefficients independently of the discarded current block given (A, J, H). The comparison coupling may correlate them with that

discarded block. It preserves the same joint marginal of the new prefix and the future original blocks, as well as the original full-network marginal, but need not preserve the canonical joint law with all discarded weights. The next entropy step uses the first marginal; original-network robustness uses the second. There is no conflicting demand for independence in the comparison coupling itself.

More explicitly, the induction invariant is that the comparison law has full original-weight marginal Q and the canonical marginal of (A, X, Y). The conditional transports supply a joint law of (A, X, Y, A') with this incoming marginal and the prescribed outgoing (A', Y) marginal. Extend it by the previous conditional law of the past original weights given (A, X, Y). The extension keeps the complete original weight law unchanged, preserves all earlier computational prefixes, and establishes the same invariant for the next layer. This is ordinary disintegration on finite-dimensional standard Borel spaces.

Rank, Gram, and readout failures together have probability at most $3L\eta$. Off these failures the expected total hidden squared disturbance is at most $20Le$. Hence

$$\mathbb{P}\{\text{no abort and } \sum_{s<L} \|\eta^s\|^2/N > r^2/16\} \le 320Le/r^2 = 0.32\alpha.$$

The final readout disturbance is at most $r/16$. Thus the total disturbance norm is at most $\sqrt{17}\, r/16 < r/2$, with exceptional probability less than $0.35\alpha < \alpha$. The small-width construction has exceptional probability at most $\alpha/4$.

A fixed finite class. Pad candidates to width $M = N_0$ and take $B = \max(B_{\text{small}}, B_{\text{large}}, 1)$. Round every parameter to a fixed grid on $[-B, B]$ of mesh at most

$$h = \frac{r}{16(d + L + 1)(2M(B + 1))^L}.$$

The bounded-activation, Lipschitz layer recursion gives uniform output change at most $r/8$. Thus rounding gives the stated numerical comparison at every successfully coupled input. Writing $P = M(d + 1) + (L - 2)M^2 + M$, the grid class satisfies

$$\log |\mathcal{H}| \le P \log \left[3 + \frac{32B(d + L + 1)(2M(B + 1))^L}{r}\right].$$

Although grid precision is exponentially small in L, its number of bits is polynomial in L. All other bounds above are polynomial too. Rounding is a fixed map on candidate networks, giving the common marginal Γ_Q and proving Theorem A.1.

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